

Day 9 Problem Set

Geometry Summer Credit | Similar Triangles and Applications

Name:	Date:	Period:
-------	-------	---------

Goal

Practice proportions from similar triangle diagrams, coordinate scale factors, and real-world similarity contexts.

Support Practice

1.	Solve: $\frac{3}{8} = \frac{x}{40}$.
2.	Solve: $\frac{12}{x} = \frac{4}{9}$.
3.	Find the scale factor from 6 to 24.
4.	Dilate (5, -1) by scale factor 3.
5.	Triangle ABC is similar to triangle DEF. $AB = 5$, $DE = 20$, $BC = 7$. Find EF.

Core Practice: Similar Triangle Diagrams

6.	Triangle ABC is similar to triangle XYZ. $AB = 10$, $XY = 25$, $BC = 14$. Find YZ.
7.	Triangle DEF is similar to triangle JKL. $DE = 8$, $JK = 20$, $EF = 12$. Find KL.

8. Triangle RST is similar to triangle MNP. $RS = 18$, $MN = 12$, $ST = 21$. Find NP.

9. Two similar triangles have scale factor 4 from small to large. The small triangle sides are 6, 9, 11. Find the large triangle sides.

10. A large triangle is a scale copy of a smaller triangle with scale factor $\frac{1}{3}$ from large to small. The large triangle has sides 15, 24, 30. Find the small triangle sides.

11. Error analysis: A student solves similar triangles by matching the shortest side in one triangle with the longest side in the other. Explain the problem.

Day 9 Problem Set

Continued

Core Practice: Coordinate and Real-World Similarity

12. A 4-foot child casts a 5-foot shadow. A pole casts a 30-foot shadow. Find the pole height.

13. A 6-foot person casts a 9-foot shadow. A tree casts a 45-foot shadow. Find the tree height.

14. A map uses 1 inch = 20 miles. Two cities are 3.5 inches apart. Find the real distance.

15. A model is built at a scale of 1 inch = 4 feet. A wall is 8.5 inches on the model. Find the actual wall length.

16. A photo is 5 inches by 7 inches. It is enlarged so the width is 15 inches. Find the new height.

17. A triangle has vertices (0,0), (3,0), (0,4). A similar triangle has vertices (0,0), (9,0), (0,12). Find the scale factor.

18. A point P(2, -6) is dilated to P'(5, -15). Find the scale factor.

19. Solve: $\frac{9}{15} = \frac{x}{25}$.

20. Spiral: A rhombus has perimeter 52. Find each side length.

21. Challenge: A rectangle is enlarged by scale factor 2.5. What happens to perimeter? What

	happens to area?
--	------------------